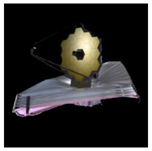

8851 - Warm matter around long-period pulsars

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
NIRCam imaging survey				
	1		NIRCam Imaging	(1) RXJ0420
	2		NIRCam Imaging	(3) RXJ1308
	3		NIRCam Imaging	(4) RXJ1605
	4		NIRCam Imaging	(5) RXJ1856
	5		NIRCam Imaging	(6) RXJ2143
	6		NIRCam Imaging	(2) RXJ0720

ABSTRACT

Disks formed around neutron stars after supernova explosions are of great astrophysical interest. Their occurrence rate and properties are essential to establish the evolutionary links between different neutron star populations, verify predictions of supernova explosion models, and enable studies of planet formation in extreme environments. According to theoretical models, such disks should be numerous around slowly rotating isolated neutron stars. X-ray irradiation from the neutron stars is expected to heat the disks which become (N)IR emitters. JWST NIRCam provides the necessary superb sensitivity and angular resolution to finally find and spatially resolve such warm circumpulsar matter. The proposed distance-limited (<500 pc) sample of most-promising disk hosts provides the best opportunity to study circumpulsar disks in detail or obtain important stringent limits on their occurrence rate.

OBSERVING DESCRIPTION

This survey of pulsars uses NIRCam imaging, aiming to detect emission at the pulsar location and potential surrounding extended emission around it.
All NIRCam observations use the SHALLOW4 readout,
The broad SW filter 150W2 and the LW filter F444W are chosen to maximize sensitivity and spectral coverage.

We use a subpixel dither standard pattern with 3 exposures.

6 targets will be observed. One target (RXJ0420) has a factor 4 longer exposure time than the other 5 targets.

For 4 targets we added PA ranges to minimize the chances of Claws scattering features. These PA ranges correspond to the **** Ranges Under 0.2 of mean **** output obtained from the JWST Rogue Path Tool in June 2025.

Proposal 8851 - Targets - Warm matter around long-period pulsars

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	RXJ0420	RA: 04 20 1.9500 (65.0081250d) Dec: -50 22 48.10 (-50.38003d) Equinox: J2000		
Comments: Category=Star Description=[Neutron stars]				
(2)	RXJ0720	RA: 07 20 24.9610 (110.1040042d) Dec: -31 25 50.21 (-31.43061d) Equinox: J2000	Proper Motion RA: -93.9 mas/yr Proper Motion Dec: 52.8 mas/yr Epoch of Position: 2001.5	
Comments: Category=Star Description=[Neutron stars]				
(3)	RXJ1308	RA: 13 08 48.2700 (197.2011250d) Dec: +21 27 6.78 (21.45188d) Equinox: J2000		
Comments: Category=Star Description=[Neutron stars]				
(4)	RXJ1605	RA: 16 05 18.5200 (241.3271667d) Dec: +32 49 18.00 (32.82167d) Equinox: J2000	Proper Motion RA: -25 mas/yr Proper Motion Dec: 142 mas/yr Epoch of Position: 2001.6	
Comments: Category=Star Description=[Neutron stars]				
(5)	RXJ1856	RA: 18 56 35.4100 (284.1475417d) Dec: -37 54 35.80 (-37.90994d) Equinox: J2000	Proper Motion RA: +325.86 mas/yr Proper Motion Dec: -59.22 mas/yr Epoch of Position: 2003.52	
Comments: Category=Star Description=[Neutron stars]				
(6)	RXJ2143	RA: 21 43 3.3800 (325.7640833d) Dec: +06 54 17.53 (6.90487d) Equinox: J2000		
Comments: Category=Star Description=[Neutron stars]				

Observation

Proposal 8851, Observation 1

Mon Jul 07 18:00:41 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Comments: center of B3 (table 3 NIRCam Apertures)

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	RXJ0420	RA: 04 20 1.9500 (65.0081250d) Dec: -50 22 48.10 (-50.38003d) Equinox: J2000		

Comments:
Category=Star
Description=[Neutron stars]

Template

Module	Subarray
B	FULL

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	NONE		STANDARD		3

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F150W2	F444W	SHALLOW4	9	2	6	3	2866.718	

Special Requirements

Offset -29.0 arcsec, 40.0 arcsec

Observation

Proposal 8851, Observation 2

Mon Jul 07 18:00:42 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Comments: center of B3 (table 3 NIRCam Apertures)

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	RXJ1308	RA: 13 08 48.2700 (197.2011250d) Dec: +21 27 6.78 (21.45188d) Equinox: J2000		

Comments:
Category=Star
Description=[Neutron stars]

Template

Module	Subarray
B	FULL

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	NONE		STANDARD		3

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F150W2	F444W	SHALLOW4	5	1	3	3	773.047	

Special Requirements

Offset -29.0 arcsec, 40.0 arcsec

Observation

Proposal 8851, Observation 3

Mon Jul 07 18:00:42 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Comments: center of B3 (table 3 NIRCam Apertures)

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	RXJ1605	RA: 16 05 18.5200 (241.3271667d)	Proper Motion RA: -25 mas/yr	
		Dec: +32 49 18.00 (32.82167d)	Proper Motion Dec: 142 mas/yr	
		Equinox: J2000	Epoch of Position: 2001.6	

Comments:
Category=Star
Description=[Neutron stars]

Template

Module	Subarray
B	FULL

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	NONE		STANDARD		3

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F150W2	F444W	SHALLOW4	5	1	3	3	773.047	

Special Requirements

Aperture PA Range 30.05262691 to 38.05262691 Degrees (V3 30.0 to 38.0)
Aperture PA Range 43.05262691 to 93.05262691 Degrees (V3 43.0 to 93.0)
Aperture PA Range 99.05262691 to 202.05262691 Degrees (V3 99.0 to 202.0)
Aperture PA Range 221.05262691 to 262.05262691 Degrees (V3 221.0 to 262.0)
Aperture PA Range 271.05262691 to 310.05262691 Degrees (V3 271.0 to 310.0)
Aperture PA Range 312.05262691 to 335.05262691 Degrees (V3 312.0 to 335.0)
Aperture PA Range 343.05262691 to 9.05262691 Degrees (V3 343.0 to 9.0)
Offset -29.0 arcsec, 40.0 arcsec

Observation

Proposal 8851, Observation 4

Mon Jul 07 18:00:42 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Comments: center of B3 (table 3 NIRCam Apertures)

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	RXJ1856	RA: 18 56 35.4100 (284.1475417d)	Proper Motion RA: +325.86 mas/yr	
		Dec: -37 54 35.80 (-37.90994d)	Proper Motion Dec: -59.22 mas/yr	
		Equinox: J2000	Epoch of Position: 2003.52	

Comments:
Category=Star
Description=[Neutron stars]

Template

Module	Subarray
B	FULL

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	NONE		STANDARD		3

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F150W2	F444W	SHALLOW4	5	1	3	3	773.047	

Special Requirements

Aperture PA Range 9.05262691 to 35.05262691 Degrees (V3 9.0 to 35.0)
Aperture PA Range 46.05262691 to 50.05262691 Degrees (V3 46.0 to 50.0)
Aperture PA Range 52.05262691 to 99.05262691 Degrees (V3 52.0 to 99.0)
Aperture PA Range 108.05262691 to 138.05262691 Degrees (V3 108.0 to 138.0)
Aperture PA Range 143.05262691 to 191.05262691 Degrees (V3 143.0 to 191.0)
Aperture PA Range 195.05262691 to 218.05262691 Degrees (V3 195.0 to 218.0)
Aperture PA Range 230.05262691 to 240.05262691 Degrees (V3 230.0 to 240.0)
Aperture PA Range 251.05262691 to 253.05262691 Degrees (V3 251.0 to 253.0)
Aperture PA Range 266.05262691 to 318.05262691 Degrees (V3 266.0 to 318.0)
Aperture PA Range 340.05262691 to 358.05262691 Degrees (V3 340.0 to 358.0)
Offset -29.0 arcsec, 40.0 arcsec

Observation

Proposal 8851, Observation 5

Mon Jul 07 18:00:42 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Comments: center of B3 (table 3 NIRCam Apertures)

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(6)	RXJ2143	RA: 21 43 3.3800 (325.7640833d) Dec: +06 54 17.53 (6.90487d) Equinox: J2000		

Comments:
Category=Star
Description=[Neutron stars]

Template

Module	Subarray
B	FULL

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	NONE		STANDARD		3

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F150W2	F444W	SHALLOW4	5	1	3	3	773.047	

Special Requirements

Aperture PA Range 36.05262691 to 29.05262691 Degrees (V3 36.0 to 29.0)
Offset -29.0 arcsec, 40.0 arcsec

Observation

Proposal 8851, Observation 6

Mon Jul 07 18:00:42 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Comments: center of B3 (table 3 NIRCam Apertures)

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	RXJ0720	RA: 07 20 24.9610 (110.1040042d)	Proper Motion RA: -93.9 mas/yr	
		Dec: -31 25 50.21 (-31.43061d)	Proper Motion Dec: 52.8 mas/yr	
		Equinox: J2000	Epoch of Position: 2001.5	

Comments:
Category=Star
Description=[Neutron stars]

Template

Module	Subarray
B	FULL

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	NONE		STANDARD		3

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F150W2	F444W	SHALLOW4	5	1	3	3	773.047	

Special Requirements

Aperture PA Range 24.05262691 to 62.05262691 Degrees (V3 24.0 to 62.0)
Aperture PA Range 66.05262691 to 98.05262691 Degrees (V3 66.0 to 98.0)
Aperture PA Range 104.05262691 to 111.05262691 Degrees (V3 104.0 to 111.0)
Aperture PA Range 123.05262691 to 261.05262691 Degrees (V3 123.0 to 261.0)
Aperture PA Range 272.05262691 to 356.05262691 Degrees (V3 272.0 to 356.0)
Aperture PA Range 358.05262691 to 14.05262691 Degrees (V3 358.0 to 14.0)
Offset -29.0 arcsec, 40.0 arcsec